

4ormIQ: Financial Decision Intelligence

What the evidence says about private exploration, conversational disclosure and decision readiness before a major financial commitment.

Research synthesis

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Peer-reviewed research, Canadian consumer data and public deployments

4orm Finance · Confidential

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Six parts, thirty-six sections and twenty-one exhibits. Every figure carries its source.

Executive summary	The chain the evidence points to, and the one question that governs it.	4
Part 1 · The hidden decision gap	Why the decision begins before the application, and what stops people entering it.	6
Part 2 · What conversation changes	Engagement, workflow, access, and the three ways conversation can fail.	13
Part 3 · Proof from deployment	Healthcare, banking and personal finance, and the boundary of what they prove.	21
Part 4 · Financial Decision Intelligence	The category, the decision-data layer, affordability, scenarios and the record.	31
Part 5 · Safety, autonomy and security	Trusted Neutrality, permissioned sharing and Canadian privacy requirements.	39
Part 6 · The research agenda	Three hypotheses, a pilot design, the kill tests and the whitespace.	43
Conclusion	What the research justifies, and what it does not.	48
Appendix · Method and evidence grading	How the evidence was weighed, and the commercial disclosure.	49
Sources	Forty-three references, each linked from the page that cites it.	50

A reference in square brackets is a link. Selecting it opens the source entry at the back of the paper.

The thirty-six sections.

01 The decision starts before the application	7	19 NatWest: from service chatbot to money understanding	27
02 Financial anxiety, shame and avoidance	8	20 Intuit and Credit Karma: personalized explanation changes behaviour	28
03 Canadians are open to digital help	9	21 Retail investors already use generative tools	29
04 The disclosure effect predates modern AI	10	22 What deployment evidence proves, and what it does not	30
05 AI can lower fear of judgment, but not consistently	11	23 The category: intelligence before the recommendation	32
06 Social privacy and technical privacy pull in opposite directions	12	24 A new decision-data layer	33
07 Conversation can increase engagement with complex material	14	25 Affordability should have four views	34
08 Front-loading context can change professional workflow	15	26 Future scenarios make trade-offs visible	35
09 A private digital front door can change who enters	16	27 Decision drift, contradiction detection and rehearsal	36
10 What people say when the interaction works	17	28 The consumer-approved decision brief	37
11 When AI feels judgmental, disclosure falls	18	29 The record can preserve before, during and after	38
12 More human-like is not always more trusted	19	30 Trusted Neutrality	40
13 Conversation can create dangerous overtrust	20	31 Private first, permissioned sharing	41
14 The evidence is spread across three sectors	22	32 Human accountability and Canadian privacy requirements	42
15 Limbic: private intake before professional assessment	23	33 The three hypotheses 4ormlQ should test	44
16 Wysa: anonymous support and institutional deployment	24	34 Proposed pilot design and measures	45
17 Headspace and Eleos: reflection plus professional intelligence	25	35 Kill tests: what would disprove the thesis	46
18 Bank of America: conversational finance at massive scale	26	36 Strategic implications and market whitespace	47

A private decision-readiness layer, not a chatbot for finance.

The opportunity is larger than "an AI chatbot for finance." The evidence points toward a private decision-readiness layer that helps consumers explore a major financial choice before the professional recommendation begins.

The potential mechanism is a chain: lower social pressure can enable more candid exploration; richer context can improve preparation; structured information can make the professional conversation more focused; and a permissioned record can preserve what the consumer understood, what changed and why the decision was made. The chain is plausible, but its combined effect in Canadian financial services has not yet been proven.

The central question

Can a secure, consumer-controlled conversational environment help a person disclose more relevant context, understand present and future affordability, identify missing questions and arrive at a regulated professional conversation better prepared, without increasing overreliance, pressure or inappropriate disclosure?

Scope. This paper synthesizes peer-reviewed disclosure research, behavioural finance, Canadian consumer research, AI privacy guidance and real-world deployments in mental health and financial services. Healthcare evidence is used as an adjacent workflow analogue, not as proof of financial outcomes. Supplier case studies are clearly distinguished from independent or peer-reviewed evidence. 4orm is developing a commercial product in this area and has an interest in the thesis.

Five signals from the evidence.

EXHIBIT 1 • SIGNALS FROM THE EVIDENCE

SIGNAL	EVIDENCE	IMPLICATION FOR 4ORMIQ
Financial avoidance	High-financial-anxiety participants were 2.75x more likely to avoid a financial task. [7]	A lower-pressure first step may matter before formal discovery.
Embarrassment	37% of Canadians agreed AI chatbots can help avoid embarrassing discussions with bank representatives. [14]	Private exploration could surface questions or facts people hesitate to say aloud.
Engagement	A conversational GenAI intervention produced 2.4x more opens and 3.8x longer engagement than static material in a 540-person RCT. [20]	Conversation can make complex material easier to work through.
Professional workflow	In a 64,862 -patient observational study, AI-supported intake was associated with 12.7 fewer assessment minutes and 45% fewer treatment-allocation changes. [19]	Front-loaded context may release professional time and reduce rework.
Safety boundary	73.4% accepted a deliberately wrong robo-adviser recommendation with a conversational interface vs 40.0% without. [11]	Understanding and persuasion must be separated.

Sources as cited. Each reference links to the source entry at the back.

01 · THE HIDDEN DECISION GAP

The decision begins before the application.

2.75x more likely to avoid a financial task, among high-financial-anxiety participants

Money carries status, identity and memory. The emotional layer decides whether a consumer seeks advice at all, looks at their own information, or discloses the facts that would help a professional understand the decision.

KEY TAKEAWAYS

- Qualification and affordability are not the same question.
- Removing a human observer changes reported behaviour, and that effect predates modern systems.
- Low social judgment helps only when technical trust is also present.

Source: financial anxiety and avoidance study [7]. Full reference at the back.

SECTION 01

The decision starts before the application.

A major financial decision starts before a consumer enters a lender, broker, dealer, adviser or insurer workflow. The person is already balancing goals, fears, obligations and future plans. Traditional systems capture the transaction once the consumer is ready to apply; they are weaker at capturing the private reasoning that shaped the decision before the application.

Qualification and affordability are not the same question

A consumer can qualify for a payment and still consider it uncomfortable. They may be planning parental leave, supporting a family member, replacing a vehicle, protecting an emergency reserve or relying on overtime that may not continue. Those circumstances can be central to the consumer's decision even when they do not alter the formal qualification calculation.

EXHIBIT 2 · FINANCIAL WELL-BEING ALREADY INCLUDES THE PRESENT AND THE FUTURE

DIMENSION	PRESENT	FUTURE
Security	Control over day-to-day and month-to-month finances	Capacity to absorb a financial shock
Freedom of choice	Ability to make choices that allow the person to enjoy life	Being on track to meet financial goals

Source: Consumer Financial Protection Bureau financial well-being framework [38]. FCAC uses a closely related definition of financial well-being in Canada [39].

The opportunity is decision readiness

The proposed 4ormIQ role is not to make the regulated decision. It is to help the consumer arrive able to explain what they need, what they can realistically carry, what could change the answer and what still requires professional judgment.

NEED

What am I trying to accomplish?

CAPACITY

What can I carry now?

FUTURE

What could change?

QUESTIONS

What still needs a professional?

— SECTION 02

Money is not a neutral data-entry exercise.

Money carries status, identity, competence, family expectations and memories of past mistakes. That emotional layer can affect whether consumers seek advice, look at their own financial information or disclose the facts that would help a professional understand the decision.

2.75x

MORE LIKELY TO AVOID A FINANCIAL TASK

9,110

PARTICIPANTS IN THE SHAME STUDIES

6

EXPERIMENTAL, ARCHIVAL AND LONGITUDINAL STUDIES

Source: Math Anxiety and Financial Anxiety Predicting Individuals' Financial Management Behavior: Study 1 n=186; Study 2 n=103. High-anxiety participants were 2.75x more likely to choose financial avoidance. [\[7\]](#)

Shame can become a financial feedback loop

Across six experimental, archival and longitudinal studies involving **9,110** participants, Gladstone and colleagues found evidence that shame can prompt financial withdrawal and disengagement, increasing the probability of counterproductive financial decisions. Shame was more strongly associated with financial hardship than guilt because it produced more withdrawal. [\[6\]](#)

The professional meeting can itself be a barrier

Gerrans and Hershey introduced "financial adviser anxiety" as concern about meeting a financial adviser. Their framework identifies embarrassment, worry and other apprehension as potential barriers to seeking professional advice. [\[8\]](#)

The challenge is not simply that people lack financial information. Sometimes the person most in need of help is also the person most uncomfortable entering the conversation.

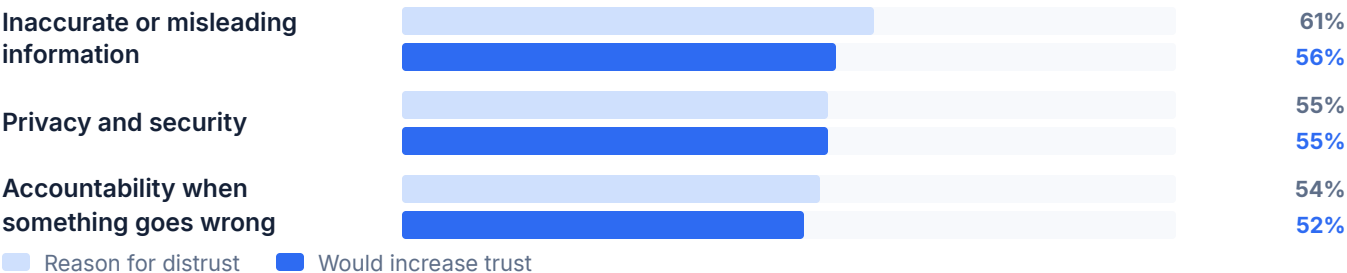
4orm synthesis of [\[6\]](#) to [\[8\]](#)

SECTION 03

Open to digital help, human for high stakes.

The 2026 TD AI Insights Report, conducted by Ipsos among **2,501** Canadian adults, shows a hybrid preference. Canadians are open to AI for budgeting and routine financial tasks, but a majority still prefer human support for financial planning and other consequential decisions. The survey's credibility interval is plus or minus 2.4 percentage points, 19 times out of 20. [\[14\]](#)

EXHIBIT 3 · THE TRUST-BUILDERS MIRROR THE CONCERNS



Source: Ipsos / TD AI Insights Report, February 2026 fieldwork, n=2,501 Canadian adults. [\[14\]](#)

A direct signal for 4ormIQ

The same TD report found that **37%** agreed that AI chatbots can help them avoid embarrassing discussions with bank representatives. That is not evidence that those conversations become deeper or more accurate, but it is a direct Canadian signal that the social barrier exists and that some consumers see a digital channel as a way around it. [\[14\]](#)

SECTION 04

The disclosure effect predates modern systems.

The idea that people answer sensitive questions differently when another human is removed from the interaction is not new. A large body of survey-method research predates generative AI and provides a stronger foundation than relying on chatbot anecdotes.

48INDEPENDENT
SAMPLES**125,672**

PARTICIPANTS

460

EFFECT SIZES

1.19CORRECTED ODDS
RATIO

Source: Gnambs and Kaspar meta-analysis: 48 independent samples, 125,672 participants, 460 effect sizes; corrected overall odds ratio 1.19. [\[2\]](#)

Computerized self-administered surveys produced significantly more reporting of socially undesirable behaviours than comparable paper-and-pencil surveys. The effect was strongest for highly sensitive topics and when surveys were administered individually. The authors caution that most samples were young and U.S.-based, and effects varied by behaviour. [\[2\]](#)

Virtual interviewers can reduce evaluation fears

Lucas and colleagues experimentally changed only how participants believed a virtual health interviewer was controlled. When participants believed the interviewer was computer-controlled, they reported lower fear of disclosure and impression management and showed greater objectively rated disclosure and more visible sadness than participants who believed a human was operating it. [\[1\]](#)

The relevant mechanism is not "AI magic." It is the change in social context when the consumer no longer feels they are performing for another person.

4orm interpretation of [\[1\]](#) to [\[2\]](#)

— SECTION 05

Lower fear of judgment, but not consistently.

A 2026 Technology in Society paper tested disclosure intention across five vignette-based experiments with **1,461** Chinese online participants. Participants reported higher intention to disclose private information to an AI psychotherapist than to either face-to-face or online human psychotherapists. Reduced fear of negative evaluation statistically mediated the effect. [\[3\]](#)

N = 1,461

Five vignette-based experiments

MEDIATOR

Fear of negative evaluation mediated the effect

BOUNDARY

Privacy-policy disclosure weakened the advantage

The result is useful, but narrow

The study measured intended disclosure in hypothetical psychotherapy scenarios, not actual mortgage, investment or lending behaviour. The participants were drawn from Chinese online samples. It supports a mechanism worth testing; it does not establish that Canadian consumers will disclose more financial information to 4ormIQ.

A 2026 health study found the opposite

Alsaad and colleagues compared face-to-face, human-through-computer and chatbot consultations in a 373-person experiment. Willingness to disclose sensitive health information was highest face-to-face (mean 4.78), lower through a human computer-mediated interaction (4.23), and lowest with the chatbot (3.85). The authors found no evidence that the chatbot created greater machine-mediated disinhibition. [\[4\]](#)

The synthesis

Low social judgment can help, but technical trust can overwhelm the benefit. The design question is therefore not "make it a bot." It is "create a private, competent and trustworthy environment where the consumer understands the boundaries."

SECTION 06

Social privacy and technical privacy pull in opposite directions.

Sensitive disclosure has at least two privacy risks. Social privacy risk is the fear that another person will judge the consumer. Technical privacy risk is the fear that the system will collect, retain, misuse or expose the information. A successful financial decision environment has to reduce both.

EXHIBIT 4 · THE DISCLOSURE EQUATION IS CONDITIONAL

CONDITION	LIKELY EFFECT	DESIGN IMPLICATION
Lower social judgment, strong technical trust	More room for candid exploration	Private-first experience; neutral language; clear purpose
Lower social judgment, weak technical trust	Consumer may disclose less or disengage	Explain collection, retention and sharing at the moment it matters
High conversational warmth, commercial conflict	Trust can become persuasion	Separate decision support from sales optimisation
Fast conclusions from thin data	AI can feel judgmental	Show what is known, unknown and assumed before suggesting next steps

Source: 4orm synthesis of [1] to [11].

A finance-specific 2025 study across three experiments (N=3,346) found that placing financial support tools inside personal customer portals increased perceived privacy risk, but also increased positive affect enough to increase willingness to disclose personal financial information. The result shows that privacy perceptions and emotional context operate together. [9]

The 2026 AI psychotherapist study found a related paradox: proactively presenting a privacy-security policy weakened the AI disclosure advantage, apparently because it made data-privacy risk more salient. The correct interpretation is not to hide privacy information. It is to make privacy comprehensible, contextual and controllable. [3]

02 · WHAT CONVERSATION CHANGES

Sustained thinking, not instant answers.

3.8x longer engagement with conversational material than with static material, in a 540-person randomized trial

The strongest benefit of a conversational system may not be that it knows more. It may be that consumers are more willing to keep working through complex material when the system responds to their specific questions, circumstances and misunderstandings.

KEY TAKEAWAYS

- Front-loaded context changed professional workflow in a **64,862**-patient study.
- The design of the first doorway can change who is willing to enter.
- Three failure modes are already documented: perceived judgment, false intimacy and overtrust.

Source: conversational engagement randomized controlled trial [20]. Full reference at the back.

SECTION 07

The advantage may be sustained thinking.

The strongest benefit of a conversational system may not be that it knows more. It may be that consumers are more willing to keep working through complex material when the system responds to their specific questions, circumstances and misunderstandings.

EXHIBIT 5 · CONVERSATIONAL AGAINST STATIC MATERIAL, SIX-WEEK RANDOMIZED TRIAL

Times opened		24 of 40	2.4x more opens than static material
Time engaged		38 of 40	3.8x longer engagement

Source: six-week randomized controlled trial, n=540, comparing a conversational tool with static CBT workbooks that delivered comparable therapeutic content. [\[20\]](#)

Participants in the conversational GenAI condition opened the material 2.4 times more often and spent 3.8 times longer engaging with it. Both groups showed comparable overall symptom reduction and equivalent safety profiles; the correct conclusion is greater engagement, not 3.8 times greater therapeutic efficacy. [\[20\]](#)

Why this matters in finance

Financial education often arrives as disclosures, calculators, articles and forms. The 4ormIQ hypothesis is that a consumer may be willing to spend twenty minutes exploring a mortgage, vehicle purchase or investment conversationally even when they would not read twenty minutes of static material. That needs direct testing, but the engagement mechanism is credible.

The product advantage may be sustained thinking, not instant answers.

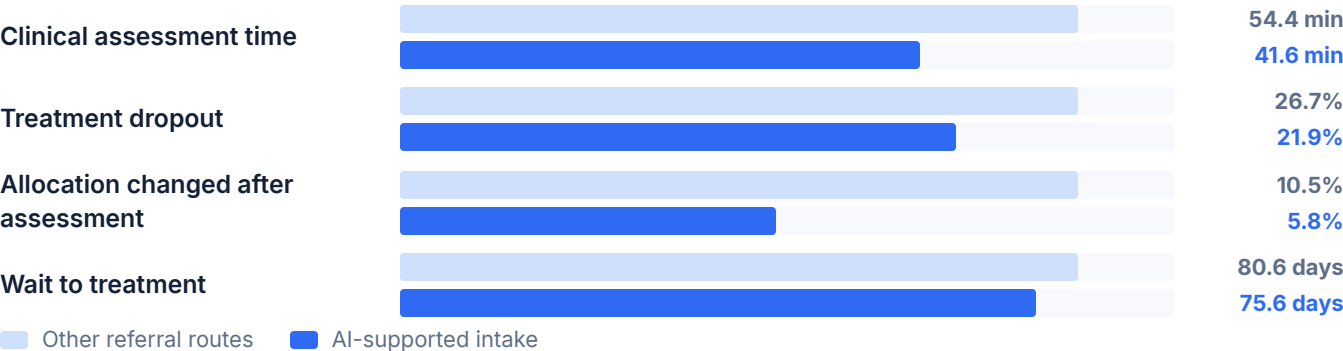
4orm synthesis

SECTION 08

Front-loading context can change professional workflow.

The closest real-world analogue to the proposed 4ormIQ workflow is not an AI therapist. It is AI-supported intake before a human professional assessment. Limbic Access collects structured clinical context before the appointment and passes it into the professional workflow.

EXHIBIT 6 · AI-SUPPORTED INTAKE IN A 64,862-PATIENT OBSERVATIONAL STUDY



Source: Rollwage et al., JMIR AI observational study across nine NHS Talking Therapies services, n=64,862; Limbic-funded, with employees among the authors. [\[19\]](#) Independent NICE evidence review also cites the 45% reduction in treatment step-ups and step-downs. [\[21\]](#)

The paper identified the provision of clinically relevant information ahead of assessment as a critical mechanism associated with the efficiency gains. This does not prove that financial discovery will save equivalent time. It establishes a workflow pattern worth testing: gather context before the professional meeting, structure it, and let the professional begin at a higher level.

The financial analogue

Instead of spending the first professional meeting discovering that income is variable, childcare begins next year or the consumer refuses to spend emergency savings, the professional could receive those consumer-approved facts before giving advice.

— SECTION 09

A private front door can change who enters the system.

In a Nature Medicine observational study covering **129,400** patients across 28 NHS Talking Therapies services, services using an AI-enabled self-referral chatbot experienced a **15%** increase in referrals, compared with **6%** in matched control services. [\[18\]](#)

+179%

REFERRALS FROM NON-BINARY INDIVIDUALS

+29%

REFERRALS FROM ETHNIC-MINORITY INDIVIDUALS

42,332

QUALITATIVE FEEDBACK RECORDS ANALYSED

Source: Habicht et al., Nature Medicine, 129,400 patients; qualitative feedback from 42,332 individuals. [\[18\]](#)

The researchers identified the human-free nature of the tool and users' self-realization of their need for support as potential drivers of improved access. These are healthcare results, and the subgroup increases should not be generalized to finance. The transferable insight is that the design of the first doorway can change who is willing to enter. [\[18\]](#)

Another deployment shows after-hours use

An Essex case study reports more than **6,500** self-referrals supported by Limbic Access, with **56%** of all self-referrals using the tool and more than **2,450** occurring outside office hours. Total referrals increased **6.5%** year over year. These are supplier and customer case metrics rather than independent causal estimates. [\[42\]](#)

— SECTION 10

What people say when the interaction works.

Testimonials are not representative evidence. They are useful for identifying the human benefit people themselves describe. Across several products, the recurring words are not "algorithm" or "automation." They are easier, anonymous, not judged, heard, understood and available when needed.

A staggered approach via a text chat helped a lot. It was a lot less overwhelming than a form.

Limbic / Essex patient testimonial [\[42\]](#)

I would have struggled to articulate myself properly to a stranger over the phone.

Limbic / Essex patient testimonial [\[42\]](#)

You can be yourself with Wysa because it's anonymous.

Wysa youth testimonial; company-selected, not representative [\[43\]](#)

Excellent and informative responses that better helped me understand my personal financial position.

Credit Karma member testimonial; company-selected, not representative [\[33\]](#)

Headspace reports that **64%** of surveyed Ebb users said they felt heard and understood after more than one million messages had been exchanged. The company also has users who publicly reject or dislike AI companionship. The signal is not universal preference; it is that a meaningful subset finds conversational reflection valuable while others want a different channel. [\[25\]](#)

— SECTION 11

When the system feels judgmental, disclosure falls.

It is tempting to assume a machine cannot judge. Consumers do not necessarily experience it that way. A 2025 MIS Quarterly paper conducted four experiments and a qualitative study and found that participants consistently perceived a text chatbot as more judgmental than a human mental-health professional even when the interactions were identical. [\[5\]](#)

Participants described the chatbot as rushing to judgment without fully understanding their situation. That perception reduced willingness to use the service, disclose information and follow recommendations. [\[5\]](#)

EXHIBIT 7 · THE DESIGN IMPLICATION

DO NOT	INSTEAD
Reach a conclusion after a short intake	State what the system understands and what it still does not know
Present an inference as a fact	Label assumptions and ask the consumer to confirm or correct them
Force a path because the model is confident	Offer exploration, skip and professional-review options
Use a diagnostic tone	Use neutral, concrete language that preserves agency

Source: four experiments and one qualitative study, MIS Quarterly 2025. [\[5\]](#)

Here is what I understand so far. Here is what is still uncertain. Would you like to explore that before we continue?

Illustrative 4ormIQ interaction

SECTION 12

More human-like is not always more trusted.

A finance-specific experiment with 410 participants compared two chatbot vignettes: Emma, designed to be more socio-emotional and human-like, and XRO23, designed to be technical, secure and reliable. The more human-like version increased perceived social presence, but did not increase trust or reduce privacy concerns. [\[10\]](#)

The unexpected disclosure result

Participants were significantly more willing to disclose several financially sensitive data items, account number, sort code and payment information, to the technical XRO23 than to the more humanized Emma. The result does not mean financial systems should be cold. It suggests that visible competence and clear system identity can matter more than simulated friendship in sensitive finance contexts. [\[10\]](#)

EXHIBIT 8 · A PROPOSED 4ORM DESIGN PRINCIPLE: TRUSTED NEUTRALITY

TOO COLD	TRUSTED NEUTRALITY	TOO HUMAN
Feels mechanical; may reduce engagement	Warm enough to feel safe; neutral enough not to manipulate; competent enough to trust; clearly a system	Can create false intimacy, blurred accountability or pressure to reciprocate

Source: 4orm working concept, grounded in [\[5\]](#), [\[10\]](#) and [\[11\]](#).

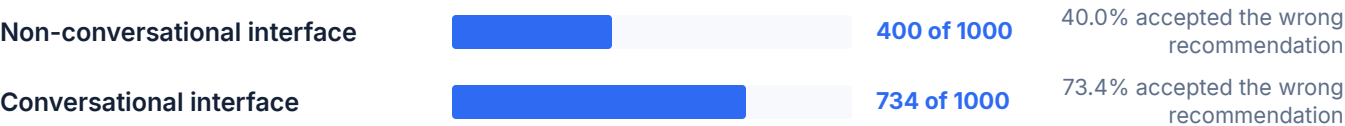
"Trusted Neutrality" is a 4orm working concept, not an established academic term. It reflects the combined evidence that consumers can benefit from low social judgment while still needing clear technical competence, boundaries and accountability.

SECTION 13

Conversation can create dangerous overtrust.

Conversational interfaces can improve trust in a financial firm. They can also make consumers more likely to accept advice they should reject. That is the clearest safety warning in this research.

EXHIBIT 9 · ACCEPTANCE OF A DELIBERATELY INCORRECT PORTFOLIO RECOMMENDATION



Source: Journal of the Academy of Marketing Science experiment comparing conversational and non-conversational robo-adviser interfaces. The portfolio recommendation was deliberately inconsistent with the participant's stated risk profile. [\[11\]](#)

Acceptance of the objectively incorrect recommendation rose from **40.0%** in the non-conversational condition to **73.4%** in the conversational condition. The effect persisted among both risk-averse and risk-seeking participants. [\[11\]](#)

The product boundary

If 4ormIQ uses deeper disclosure to increase product conversion, the architecture becomes fundamentally different. The purpose of consumer intelligence should be better understanding, not greater compliance with a commercial recommendation.

The more persuasive the interface becomes, the more important it is to separate understanding from selling.

4orm synthesis of [\[11\]](#)

03 · PROOF FROM DEPLOYMENT

No single company proves the whole thesis.

129,400 patients across 28 NHS Talking
Therapies services in the access study

The evidence exists in pieces. Healthcare companies show private intake, structured context and professional handoff. Banks show massive consumer adoption of conversational finance. Personal-finance platforms show personalized explanation and commercial effects.

KEY TAKEAWAYS

- The opportunity is combining those mechanisms while preserving consumer control.
- Supplier-funded results are marked as such throughout, and independent reviews are cited where they exist.
- None of it establishes a causal improvement to suitability or affordability in Canadian finance.

Source: Nature Medicine access study [18]. Full reference at the back.

SECTION 14

The evidence is spread across three sectors.

No single company currently proves the full 4ormIQ thesis. The evidence exists in pieces. Healthcare companies show private intake, structured context and professional handoff. Banks show massive consumer adoption of conversational finance. Personal-finance platforms show personalized explanation and commercial effects. The opportunity is in combining those mechanisms while preserving consumer control.

EXHIBIT 10 • DEPLOYMENT LANDSCAPE

COMPANY OR STUDY	WHAT IT DEMONSTRATES	EVIDENCE STRENGTH	TRANSFERABLE LESSON
Limbic	AI-enabled intake before human assessment; access, efficiency and engagement	Peer-reviewed observational and RCT; some company-funded	Understand the person first; hand structured context to the professional
Wysa	Anonymous conversational support, referral and enterprise deployments	Peer-reviewed work plus supplier cases	Anonymity and on-demand access can reduce barriers
Headspace Ebb	Conversational reflection at consumer scale	Company usage and feedback	Reflection itself can be a useful product layer
Eleos	Conversation-derived professional documentation and intelligence	Customer case	Consumer and professional intelligence can coexist
Bank of America Erica	Large-scale conversational financial usage	Company operating data	Consumers will repeatedly converse about money
NatWest	Natural-language spending exploration; GenAI service journeys	Bank operating data	Conversation is moving from service to financial understanding
Intuit / Credit Karma	Personalized explanations and measurable engagement and conversion effects	Investor reporting plus company cases	Personalized explanation changes behaviour; conflicts must be governed

Source: as cited in sections 15 to 21. Evidence strength is stated for each row.

SECTION 15

Limbic: private intake before professional assessment.

Limbic is the strongest architectural analogue in this study because its product sits before the professional encounter. The system asks questions, collects clinically relevant information, structures the result and supports the clinician who remains responsible for care.

129,400

PATIENTS IN THE ACCESS STUDY [18]

64,862

PATIENTS IN THE EFFICIENCY STUDY [19]

540

PARTICIPANTS IN THE ENGAGEMENT TRIAL [20]

The mechanism is more important than the sector

Limbic's results do not prove that mortgage or investment discovery will behave the same way. They show that a digital system can collect useful context before a professional appointment, that people will complete the interaction, and that the downstream professional process can change when better information arrives earlier.

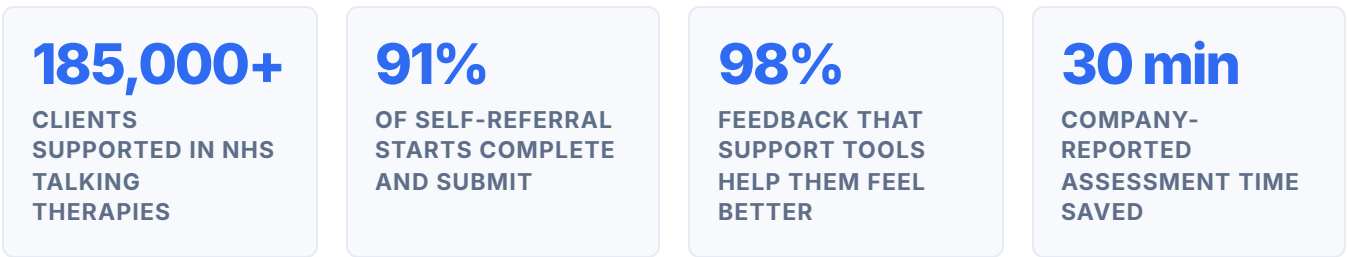
A 2026 NICE evidence review treated some Limbic results as outside its formal scope and noted evidence limitations, while still citing a statistically significant **45%** reduction in post-assessment treatment step-ups and step-downs in the **64,862**-patient dataset. [21] The presence of an independent review is useful because much of the underlying research is company-funded.

Source: conventional comparator values were 54.4 minutes assessment time, 26.7% dropout and 10.5% treatment-allocation change. [19]

SECTION 16

Wysa: anonymous support and institutional deployment.

Wysa is useful for a different reason: it demonstrates how anonymity, structured conversational support and human-in-the-loop pathways can operate at scale inside institutional programs.



Source: Wysa NHS Talking Therapies product page; company-reported deployment metrics. [22] NICE's 2026 early value assessment notes uncertainty around assessment time savings and did not find direct comparative evidence for Wysa on that outcome. [21]

EXHIBIT 11 · ENTERPRISE CASE SIGNALS

DEPLOYMENT	REPORTED RESULT	EVIDENCE BOUNDARY
Vitality, 60,000 higher-risk members offered Wysa	81% initiation-to-onboarding; 88% returned for 2 or more sessions; 83% found Wysa useful	Supplier or customer case, not randomized [23]
Travelers workers' compensation, 5,000+ employees over two years	20% uptake; 33% reduction in lost-time days; US\$21M costs saved	Supplier or customer case; causality and attribution require caution [24]
NHS pathway	Anonymous conversation, pre-populated clinical summaries and referral support	Operational deployment; performance metrics vary by service [22]

Source: supplier and customer case studies [22] to [24]. Not independent causal estimates.

The useful product lesson is not "copy Wysa." It is that some people value a stigma-free, always-available channel and that structured digital intake can connect into a human service rather than replacing it.

— SECTION 17

Reflection on one side, professional intelligence on the other.

Headspace Ebb: the consumer reflection layer

Headspace launched Ebb as a conversational emotional-reflection companion. By April 2025, U.S. members had exchanged more than one million messages; in user-feedback surveys, **64%** reported feeling heard and understood. Ebb was trained around motivational interviewing and is positioned as support for reflection rather than a replacement for clinical care. [\[25\]](#)

Eleos: the professional intelligence layer

Eleos works on the other side of the interaction. Its systems analyze care conversations and help create documentation and structured insights for professionals. In a ReachLink case study, average progress-note time fell from 12.7 to 7.3 minutes, a **42.5%** reduction; the customer also reported clinicians could see **20%** more patients and reduce administration by one to two hours per day. These are company and customer case results, not independent causal estimates. [\[26\]](#)

1M+

EBB MESSAGES BEFORE UK
ROLLOUT [25]

64%

FELT HEARD AND
UNDERSTOOD [25]

42.5%

REACHLINK PROGRESS-NOTE
TIME REDUCTION [26]

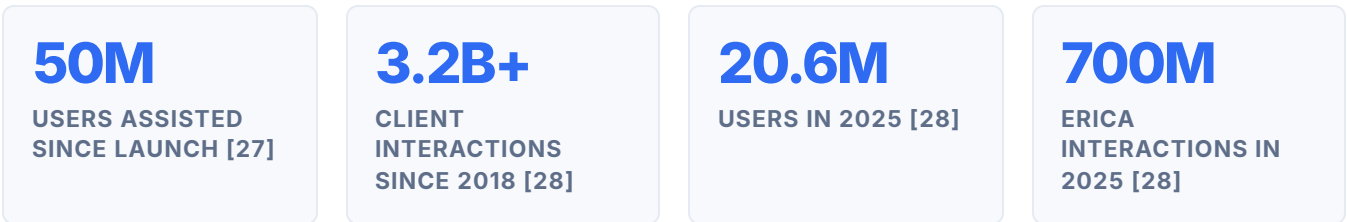
The combined architecture

4ormIQ can learn from both directions: help the consumer think privately, then help the professional receive structured context. The differentiator is the permission layer connecting the two without treating the consumer's private exploration as automatically available sales data.

SECTION 18

Consumers will repeatedly converse about money.

Bank of America's Erica demonstrates that consumers will repeatedly use a conversational financial assistant once it becomes embedded in a trusted banking environment.



In August 2025 Bank of America said Erica averaged more than **58 million** interactions per month, users had spent **18.7 million** hours conversing with the assistant, and more than **98%** found the information they needed. The bank also reported more than **1.7 billion** proactive personalized insights. [27]

EXHIBIT 12 · WHAT ERICA VALIDATES

VALIDATED	NOT VALIDATED
Consumers will repeatedly use natural-language interfaces around their money	That bank-owned assistants create deeper or more candid major-decision discovery
Conversation can move routine information and support away from human specialists	That the assistant improves long-term financial well-being
Personalized insights can become a core banking interaction	That consumers control how intimate information is reused across sales or underwriting

Source: Bank of America company operating data. [27] [28]

The whitespace for 4ormIQ is not "another bank chatbot." It is deeper, consumer-controlled exploration before a high-stakes financial commitment.

SECTION 19

From service chatbot to money understanding.

NatWest shows the evolution from service automation toward conversational financial self-understanding. Its Cora assistant handled **12.9 million** retail-banking conversations in 2025, with **50%** requiring no human intervention. GenAI was added to 21 customer journeys, producing about a 20-percentage-point uplift in queries resolved without human intervention compared with equivalent non-GenAI journeys. [\[29\]](#)



The more important direction: talk to your money

NatWest's Spending Insights experience allows customers to ask natural-language questions about spending, compare periods, interrupt, ask follow-up questions and add context. The system combines tailored insights with interactive charts and visualizations. [\[30\]](#)

EXHIBIT 13 · COMMON CONSUMER USES, NATWEST 2026 RESEARCH



Source: NatWest Group 2026, next generation tools for spending insights and fraud support. [\[31\]](#)

The shift is from "show me my transactions" toward "help me understand what my money means."

4orm interpretation of NatWest direction [\[29\]](#) to [\[31\]](#)

SECTION 20

Personalized explanation changes behaviour.

Intuit has deployed GenAI experiences at substantial scale: more than **20 million** TurboTax customers and **30 million** Credit Karma members, according to its FY25 Investor Day materials. [\[32\]](#)

EXHIBIT 14 · SELECTED INTUIT-REPORTED OUTCOMES

EXPERIENCE	REPORTED RESULT	WHAT IT SUGGESTS
Credit Karma "See Why" recommendation explanation	3% credit-card conversion lift; 90% positive feedback; 1% sitewide revenue per user lift	Personalized explanation can change commercial behaviour
TurboTax free-form GenAI responses	28% lower contact rate; 4 points higher funnel progression	Explanation can reduce human support demand
TurboTax embedded GenAI explanations	74% helpfulness; 14 points above non-GenAI users	Personalized explanation can improve perceived usefulness
Credit Karma financial monitoring	72% positive feedback	Consumers respond to data-grounded financial interpretation

Source: Intuit FY25 Investor Day. Results are company-reported; the See Why experiment was statistically significant for iOS members in July and August 2024. [\[32\]](#)

Credit Karma says members can ask questions grounded in their financial profiles, explore spending and understand trade-offs. Company-selected member testimony emphasizes the "conversational aspect" and improved understanding of personal financial position. [\[33\]](#)

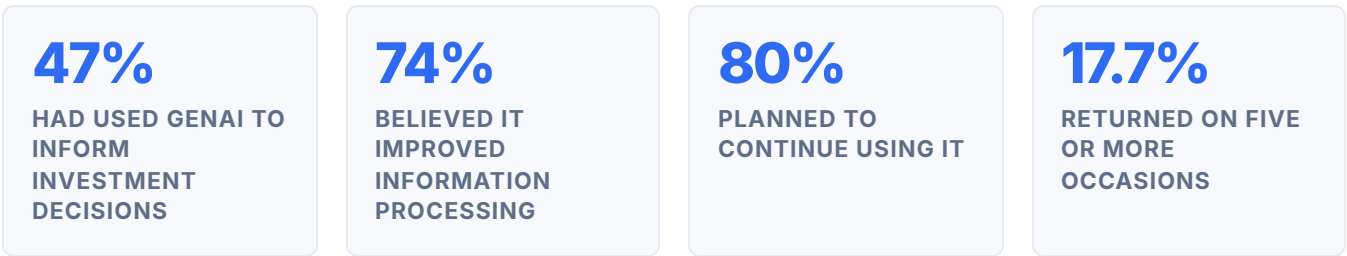
The conflict to manage

The same system that improves understanding can increase conversion. That creates a governance issue for any consumer-intelligence product: when a deeper understanding of the consumer also benefits the seller, the consumer needs visible control over purpose and sharing.

SECTION 21

Retail investors already use generative tools to interpret.

A 2026 Journal of Accounting and Economics paper analyzed **412,192** actual investor questions submitted to Alpha, Public.com's brokerage-integrated GenAI tool, and surveyed **2,175** retail investors. [\[34\]](#)



Source: Blankespoor, Croom and Grant, 2026. 412,192 investor questions; survey n=2,175. [\[34\]](#)

In the query data, repeated users shifted from broad screening toward detailed monitoring and interpretation. [\[34\]](#)

The barriers match the Canadian trust data

EXHIBIT 15 · LIMITATIONS IDENTIFIED BY RETAIL INVESTORS



Source: Blankespoor, Croom and Grant, 2026 [\[34\]](#). These concerns are closely aligned with the TD and Ipsos Canadian findings on inaccurate information, privacy and accountability. [\[14\]](#)

Consumers are already using generative systems as an information-processing layer. The question is whether that layer can be made safer, more consumer-controlled and more useful for major decisions.

4orm synthesis

SECTION 22

What the deployment evidence does not prove.

EXHIBIT 16 · EVIDENCE BOUNDARY

EVIDENCE	WHAT IT SUPPORTS	WHAT IT DOES NOT PROVE
Disclosure experiments	Social context, evaluation fear and agent identity can change willingness to disclose	That every consumer will tell 4ormIQ more than a professional
Financial anxiety and shame studies	Emotional barriers can contribute to avoidance and disengagement	That AI is the correct intervention for every financially anxious consumer
Limbic and Wysa healthcare deployments	Pre-professional digital intake can capture context and change workflow at scale	Equivalent time, access or outcome effects in mortgages, auto, insurance or investments
Bank and fintech deployments	Consumers use conversational finance at scale and value personalization	Independent improvements in long-term financial well-being
Robo-adviser experiment	Conversation can materially increase trust and recommendation acceptance	That conversational interfaces are inherently unsafe; rather, they require safeguards
Canadian surveys	Consumers are open to AI support and value privacy, accountability and human oversight	Actual behaviour in a 4ormIQ major-decision workflow

Source: 4orm synthesis of the evidence cited throughout this paper.

The thesis remains a hypothesis until tested in finance

The proposed combination, private exploration, consumer-controlled sharing, future-scenario testing and professional handoff, has not been established as a causal improvement to suitability, affordability or consumer outcomes. The paper therefore treats 4ormIQ capabilities as a research and product direction, not a proven result.

04 · FINANCIAL DECISION INTELLIGENCE

Intelligence before the recommendation.

8 fields of decision data the transaction record does not capture

A category larger than a chatbot and narrower than automated advice: a private, consumer-controlled environment that helps a person understand a major financial decision before the professional recommendation begins.

KEY TAKEAWAYS

- Affordability is four questions, not one number.
- A future scenario is not a prediction; it is a test of whether the consumer would still want the decision.
- The transferable object is a consumer-approved brief, not the private transcript.

Source: Exhibit 18, decision data the transaction often does not capture.

SECTION 23

A category larger than a chatbot, narrower than advice.

The research suggests a category larger than a chatbot and narrower than automated advice. We use the working term Financial Decision Intelligence: a private, consumer-controlled environment that helps a person understand a major financial decision before the professional recommendation begins.

EXHIBIT 17 • FIVE JOBS FOR 4ORMIQ

JOB	CONSUMER QUESTION	SYSTEM ROLE
Explore	What am I really trying to do?	Let the person explain the situation in natural language without needing the right financial terminology.
Understand	What matters, and what am I missing?	Structure goals, constraints, uncertainties and knowledge gaps.
Stress-test	What happens if life or the product changes?	Model reasonable scenarios without pretending to predict the future.
Prepare	What do I need to ask before I commit?	Turn the conversation into questions and a consumer-readable decision brief.
Connect	What should my professional see?	Let the consumer approve a structured handoff while retaining private exploration where appropriate.

Source: 4orm product framework. Proposed capabilities, not validated outcomes.

The governing distinction

4ormIQ helps the consumer understand the decision. The regulated professional remains responsible for professional judgment, advice and obligations that attach to their role. The system can surface uncertainty; it should not convert missing information into a confident recommendation.

Explore. Understand. Stress-test. Prepare. Share.

Proposed 4ormIQ consumer flow

SECTION 24

A new decision-data layer.

Traditional financial data describes the balance sheet and transaction. A decision-readiness system can add a separate layer describing the reasoning around the transaction. These categories are proposed 4orm concepts and should be treated as product design fields, not established regulatory classifications.

EXHIBIT 18 · DECISION DATA THE TRANSACTION OFTEN DOES NOT CAPTURE

DECISION DATA	WHAT IT RECORDS	ILLUSTRATIVE EXAMPLE
Intent	The outcome the person is trying to achieve	Move before the next school year.
Constraint	A boundary that cannot or should not be ignored	Keep six months of cash untouched.
Preference	Which trade-off matters more	Flexibility matters more than the lowest initial payment.
Scenario	A plausible future condition that changes the picture	Parental leave within 18 months.
Understanding	What the consumer can explain, and what remains unclear	Understands renewal risk; unsure about prepayment penalty.
Confidence	Where the consumer is certain versus hesitant	Comfortable with payment; uncertain about closing-cost reserve.
Change	How priorities or facts changed during the journey	Original payment boundary increased after viewing a specific property.
Permission	What the consumer has chosen to share and with whom	Professional may see the decision brief, not the private conversation transcript.

Source: 4orm product framework. Proposed design fields, not regulatory classifications.

Relevant completeness, not more data

The value is not collecting more data for its own sake. The value is relevant completeness: enough context to understand the decision, no more than necessary, with purpose and permission visible to the consumer.

SECTION 25

Affordability should have four views.

Most affordability conversations collapse different questions into one number. Financial Decision Intelligence can separate them so the consumer can see the difference between what is available and what remains comfortable under their own priorities.

EXHIBIT 19 · FOUR AFFORDABILITY VIEWS

VIEW	QUESTION	WHAT IT USES
Maximum qualification	What might the institution approve?	Formal rules, verified income, debt and product criteria.
Present affordability	What works under today's actual cash flow?	Income, recurring expenses, existing obligations and current reserves.
Comfortable affordability	What leaves room for the life the consumer says matters?	Goals, savings preferences, lifestyle commitments and self-defined boundaries.
Resilient affordability	What remains workable under reasonable future stress?	Scenario testing: rates, income change, childcare, repairs, insurance, emergencies and other material shocks.

Source: framework proposed by 4orm. CFPB and FCAC financial well-being research supports evaluating present control, future resilience, goals and freedom of choice rather than relying only on income or credit measures. [\[38\]](#) [\[39\]](#)

QUALIFY

Institutional maximum

TODAY

Present cash flow

COMFORT

Consumer boundary

RESILIENCE

Future stress

The system should not invent certainty

A future scenario is not a prediction. The safe question is not "this will happen." It is "if this happened, would you still want the decision?"

SECTION 26

Future scenarios make trade-offs visible.

Major financial commitments are long-lived. The facts a consumer knows today can be tested against plausible changes without pretending the system can predict the future.

EXHIBIT 20 · ILLUSTRATIVE SCENARIO LIBRARY

DECISION	POSSIBLE SCENARIO	QUESTION THE CONSUMER CAN EXPLORE
Mortgage	Renewal rate 2 percentage points higher	Does the payment still leave the reserve and monthly savings you said matter?
Mortgage	Income 20% lower for six months	What would you cut, and how long could reserves cover the gap?
Home purchase	Childcare \$1,800 a month higher	Does the housing payment still fit the family plan?
Automotive	Vehicle exit after three years	What balance could remain relative to estimated value?
Insurance	Premium increases or coverage changes	Would the policy still fit the household budget and protection goal?
Investment	20% market decline before a near-term goal	Would you need to sell, delay the goal or use other funds?

Source: 4orm product framework. Illustrative scenarios, not predictions.

Just-in-time information matters

FCAC's National Financial Literacy Strategy explicitly encourages behavioural design and just-in-time information at the point of decision. [13] In a separate randomized study, a five-minute digital intervention improved reported credit-card debt outcomes, with the intervention group reducing reported balances 35% more than the control group. [12]

Source: FCAC randomized study; more than 25,000 completed the initial survey, and balance-carrying participants were randomized. [12]

— SECTION 27

Drift, contradiction and rehearsal.

Decision drift

A consumer may set a boundary when thinking privately, then change it after falling in love with a house, vehicle or investment. The change may be rational. The system does not need to decide. It can preserve the earlier statement and ask what changed.

Three weeks ago you said a \$2,800 payment was your comfort boundary. You are now considering \$3,250. What changed?

Illustrative 4ormIQ prompt

Contradiction detection

Natural-language discovery can reveal tensions between stated priorities. "Security is my top priority" and "I am comfortable using almost all available savings" may both be true, but the combination deserves explicit review.

Earlier you said keeping an emergency reserve matters. This scenario leaves substantially less. Would you like to review that trade-off?

Illustrative 4ormIQ prompt

Conversation rehearsal

Some consumers know what they feel but cannot articulate it professionally. A private system can help turn "I don't want to be house poor" into a structured statement about a monthly boundary, savings goal, reserve and upcoming life change. The professional receives a clearer starting point without the system making the recommendation.

These capabilities are product hypotheses. They should be tested for comprehension, usefulness, unintended pressure and whether consumers perceive the system as accurately representing their intent.

SECTION 28

The consumer-approved decision brief.

The professional should not need the consumer's entire private conversation. The transferable object can be a consumer-readable, consumer-approved decision brief.

EXHIBIT 21 · ILLUSTRATIVE DECISION BRIEF

FIELD	EXAMPLE
Goal	Purchase a primary residence within six months.
Comfort boundary	Keep total housing payment at or below the consumer-defined comfort range.
Important constraint	Maintain six months of emergency savings after closing.
Expected change	Parental leave likely within 18 months.
Scenario concern	Consumer becomes uncomfortable under a renewal-rate scenario 2 points higher.
Open questions	Fixed versus variable; prepayment flexibility; closing-cost reserve.
What the consumer chose to share	Summary above plus selected supporting documents; private exploration transcript withheld.

Source: 4orm product framework. Illustrative content, not a live output.

The professional starts with context, not a blank page

A broker, adviser, dealer or other regulated intermediary can still verify facts, challenge assumptions and exercise judgment. The benefit is that the consumer arrives with a clearer story and the professional can spend more time on the trade-offs that require expertise.

This is the point where consumer understanding and business evidence begin to connect: the consumer controls what moves forward, while the business can preserve the information it legitimately relied on.

SECTION 29

The record can preserve before, during and after.

EXHIBIT 22 · A DECISION RECORD IS MORE THAN A SIGNED DOCUMENT

STAGE	EVIDENCE TO PRESERVE	WHY IT MATTERS
Before	Goals, circumstances, concerns, original boundaries, scenario results and consumer-approved context	Shows what the consumer was trying to accomplish before a product was selected.
During	Professional questions, clarifications, verified facts, assumptions, recommendation reasoning and unresolved items	Shows how the professional connected the person to the option.
Change	Material facts or preferences that changed and who changed them	Separates genuine preference change from unexplained drift.
Decision	What the consumer reviewed, questioned and approved	Supports comprehension and a coherent record of the choice.
After	Supporting documents, corrections, permissions, review actions and retained evidence	Makes later supervision, complaint review or audit less dependent on reconstructing scattered records.

Source: 4orm product framework.

The business benefit follows the consumer benefit

The design principle is important: 4ormIQ should not ask consumers to disclose intimate context mainly so the business can build a better file. The consumer should first receive direct value: clarity, scenarios, questions and preparation. The business benefit follows through a permissioned, structured record.

Understanding before the product. Evidence after the decision.

4orm positioning

05 • SAFETY, AUTONOMY AND SECURITY

Warm enough to feel safe. Neutral enough not to manipulate.

73.4% accepted a deliberately wrong recommendation when the interface was conversational, against 40.0% without

Security is not separate from disclosure. It is one of the conditions that makes candid exploration possible. A consumer who does not understand where their information goes may reasonably withhold it.

KEY TAKEAWAYS

- Do not pretend to be human; visible competence mattered more than simulated warmth.
- Privacy has to be visible inside the workflow, at the moment each question is asked.
- The organization, not the system, remains accountable for consequential decisions.

Source: conversational robo-adviser experiment [11]. Full reference at the back.

SECTION 30

Trusted Neutrality.

The combined research supports a design position between two extremes: a cold form and a synthetic friend. A financial decision system should be warm enough to reduce friction, neutral enough not to manipulate, competent enough to earn trust and explicit enough that the consumer never forgets it is a system.

EXHIBIT 23 · DESIGN RULES FROM THE EVIDENCE

RULE	EVIDENCE BEHIND IT
Do not pretend to be human	A more human-like finance chatbot did not increase trust and reduced willingness to disclose several sensitive financial items. [10]
Understand before concluding	Chatbots perceived as rushing to judgment were seen as more judgmental and reduced disclosure. [5]
Show uncertainty	OECD identifies hallucination, bias and response-quality risk in personal-finance AI. [15]
Keep advice boundaries visible	A conversational robo-adviser increased acceptance of an incorrect recommendation. [11]
Let the consumer choose the channel	Consumer testimony is mixed; some value anonymity while others strongly prefer human interaction. [25] [43]
Make the system useful before asking for more data	Disclosure should be purpose-limited; privacy regulators caution against indiscriminate collection. [16]

Source: as cited in each row.

Trusted Neutrality is a proposed house principle. It should be validated through user testing, including whether consumers interpret the system as unbiased, whether they understand commercial relationships and whether emotionally warm language changes their willingness to challenge the system.

SECTION 31

Private first, permissioned sharing.

Security is not separate from disclosure. It is one of the conditions that makes candid exploration possible. A consumer who does not understand where their information goes may reasonably withhold it.

EXHIBIT 24 · PRIVACY SHOULD BE VISIBLE IN THE WORKFLOW

CONSUMER QUESTION	PRODUCT ANSWER TO MAKE VISIBLE
Why are you asking me this?	Show the decision purpose beside the question.
Does my professional see this?	State whether the answer stays private, becomes part of the brief, or requires explicit sharing.
Can I skip it?	Allow optional questions to be skipped without disguising them as mandatory.
Can I correct it?	Give the consumer a readable view of inferred and supplied information and a correction path.
How long is it kept?	Show a clear retention rule appropriate to the purpose and legal requirements.
Can my vulnerability be used to sell to me?	Create governance that separates sensitive exploration from commercial optimization unless a legitimate, transparent purpose exists.

Source: 4orm product framework, aligned to Canadian privacy guidance. [\[16\]](#)

Canadian privacy regulators advise organizations using generative AI to establish legal authority, use appropriate purposes, limit collection, explain how personal information is used, provide access and correction, set retention schedules and use safeguards proportionate to sensitivity. They also state that inferences about identifiable individuals are personal information. [\[16\]](#)

Private first does not mean invisible accountability

A consumer-controlled experience still needs clear records of permissions, corrections, data sources and consequential outputs. Privacy and evidentiary integrity have to be designed together.

SECTION 32

Human accountability and Canadian privacy requirements.

The public expectation and the regulatory direction are aligned: accuracy, privacy, accountability and human oversight matter most when AI enters consequential decisions.

EXHIBIT 25 · CANADIAN PRIVACY PRINCIPLES THAT DIRECTLY AFFECT THE DESIGN

PRINCIPLE	IMPLICATION FOR 4ORMIQ
Legal authority and meaningful consent	Know and document the authority for collection, use, disclosure and deletion. Avoid deceptive design patterns.
Appropriate purpose	Collect and use personal information only for purposes a reasonable person would consider appropriate.
Openness	Explain what, how, when and why information is collected and where generated outputs matter.
Access and correction	Provide a practical way for consumers to see and correct information and inferences about them.
Limiting collection and retention	Do not collect information merely because it may be useful later; define retention schedules for prompts and outputs.
Accuracy and safeguards	Validate outputs used in high-risk contexts; protect information with safeguards proportionate to sensitivity.
Human accountability	The organization, not the automated system, remains accountable for consequential decisions and should provide human review mechanisms.

Source: Office of the Privacy Commissioner of Canada and provincial privacy authorities, Principles for responsible, trustworthy and privacy-protective generative AI technologies. [\[16\]](#) Public expectation data from TD and Ipsos 2026, n=2,501 Canadians. [\[14\]](#)

06 · THE RESEARCH AGENDA

Three hypotheses. All three must move.

3 hypotheses that must move together; if one fails, the product thesis changes

The evidence supports a serious pilot, not a victory declaration. A system that improves disclosure but weakens privacy is not a success. A system that saves professional time but makes consumers more compliant with a recommendation is not a success.

KEY TAKEAWAYS

- Mortgage discovery is the logical first pilot: large commitment, multi-year affordability, extensive discovery.
- The kill tests are defined in advance, and a single material failure blocks the thesis.
- Do not average away a blocker: satisfaction never offsets a privacy failure.

Source: Exhibit 26, the proposed 4ormIQ research model.

SECTION 33

The three hypotheses 4ormIQ should test.

The evidence supports a serious pilot, not a victory declaration. A useful study should test three hypotheses together. If one fails, the product thesis changes.

EXHIBIT 26 · THE PROPOSED 4ORMIQ RESEARCH MODEL

HYPOTHESIS	WHAT WOULD SUPPORT IT	WHAT WOULD WEAKEN IT
H1 · Consumer decision readiness	Users identify more relevant context, future constraints and unanswered questions; comprehension improves	No meaningful difference from conventional discovery, or confusion increases
H2 · Professional usefulness	Consumer-approved briefs reduce repetitive discovery, surface material issues earlier and improve record completeness	Professionals ignore the brief, spend more time correcting it or find it unreliable
H3 · Preserved autonomy	Users feel in control, understand system limits, challenge the system when needed and do not regret disclosure	Higher overreliance, perceived pressure, unwanted sharing or excessive trust

Source: 4orm research agenda.

The combined outcome matters

A system that improves disclosure but weakens privacy is not a success. A system that saves professional time but makes consumers more compliant with a recommendation is not a success. A system that feels empowering but produces incomplete or inaccurate records is not a success.

Better understanding, better professional context and preserved autonomy. All three must move in the right direction.

4orm research criterion

Proposed pilot design and measures.

A first controlled pilot could focus on one decision type. Mortgage discovery is a logical starting point because the commitment is large, affordability is multi-year and professional discovery already collects extensive consumer information.

EXHIBIT 27 · CORE MEASUREMENTS

MEASURE	HOW TO MEASURE IT	WHY IT MATTERS
Relevant context surfaced	Independent rubric comparing material facts disclosed across conditions	Tests whether the system improves the information set.
Future constraints identified	Count of verified upcoming obligations and plausible scenarios raised before the meeting	Tests whether affordability later becomes more visible.
Consumer comprehension	Explain key trade-offs in own words; scored against a pre-defined rubric	Measures understanding rather than signature or satisfaction.
Question readiness	Number and quality of unresolved questions carried into the professional meeting	Tests whether the consumer arrives prepared.
Professional discovery time	Timed professional discovery and number of repetitive questions	Tests workflow value.
Recommendation changes	Record whether surfaced information materially changes professional options or advice	Tests whether earlier context affects the actual process.
Recall at 7 and 30 days	Repeat comprehension and key-term recall	Tests whether understanding persists.
Autonomy and pressure	Validated or pre-registered scales plus qualitative interviews	Tests whether conversation becomes coercive.
Regretted disclosure	Ask whether users shared anything they wish had stayed private	Tests privacy harm.
Overreliance	Introduce a clearly questionable suggestion and measure challenge and escalation behaviour	Tests whether trust displaces judgment.

Source: 4orm proposed pilot design.

Pre-register the hypotheses, use the same decision content across conditions where possible, and separate product analytics from outcome scoring. The pilot should compare 4ormIQ against the actual current process, not against an intentionally weak control.

SECTION 35

Kill tests: what would disprove the thesis.

The research is stronger when the team defines in advance what evidence would cause it to change direction. These are proposed kill tests.

EXHIBIT 28 · A SINGLE MATERIAL FAILURE CAN BLOCK THE THESIS

KILL TEST	FAILURE SIGNAL	ACTION
Disclosure quality	Users disclose more volume but not more relevant or accurate context	Reduce conversational depth; redesign question logic.
Comprehension	Users feel more confident but cannot explain trade-offs more accurately	Treat confidence as a risk; strengthen grounding and checks.
Autonomy	Users are materially more likely to accept questionable suggestions	Remove recommendation-like language; increase friction and human review.
Privacy	Meaningful share of users regret disclosure or misunderstand who can see information	Rebuild permission and progressive-disclosure model.
Professional usefulness	Professionals spend substantial time correcting or re-asking information	Improve verification, provenance and brief structure.
Commercial conflict	Sensitive disclosures materially change marketing or sales pressure without clear consumer benefit and permission	Create technical and policy firewall between exploration and conversion.
Equity and accessibility	The experience performs materially worse for lower-literacy, older, disabled or language-diverse users	Redesign channel options and accessibility before scaling.

Source: 4orm proposed kill tests.

Do not average away a blocker

A high satisfaction score should not offset a privacy failure. A faster professional workflow should not offset worse consumer comprehension. The research program needs separate release gates for truth, safety, comprehension, autonomy and operational value.

SECTION 36

Where the whitespace appears.

Most financial technology products optimize one side of the relationship: the institution's workflow, the consumer's budgeting, a product recommendation, or a transaction. The research points toward a less developed layer: consumer-controlled understanding before a consequential financial commitment.

EXHIBIT 29 · WHERE THE WHITESPACE APPEARS

CATEGORY	PRIVATE EXPLORATION	FUTURE SCENARIOS	PROFESSIONAL HANDOFF	CONSUMER PERMISSION	DECISION EVIDENCE
Traditional application	Low	Low	Yes	Low	Medium
Budgeting or PFM app	High	Some	Usually no	Medium	Low
Bank chatbot	High	Growing	Some	Low to medium	Medium
Robo-adviser	High	Product-specific	Limited	Low	Medium
Healthcare intake analogue	High	Context-specific	High	Medium	High
Proposed 4ormIQ direction	High	High	High	High	High

Source: 4orm analysis. The 4ormIQ row describes a proposed architecture, not current validated performance.

The strategic distinction

Most incumbents ask: "How can AI help us understand our customer better?" The 4ormIQ thesis begins one step earlier: "How can intelligence help the consumer understand themselves and the decision better?" With permission, better institutional context can follow.

That ordering creates a possible trust advantage. It also creates a higher obligation: the system must visibly serve the consumer before it serves the transaction.

Not an automated adviser. A decision-readiness layer.

The research does not justify a claim that consumers always tell AI more. It shows something more useful: disclosure is sensitive to social judgment, technical trust, context and the design of the interaction.

Financial anxiety and shame can produce avoidance. Digital environments can reduce some barriers, but poorly trusted or prematurely judgmental systems can create new ones.

The strongest adjacent deployment evidence shows that conversational intake can gather useful context before a human professional encounter, change access, increase engagement and improve workflow. Financial deployments show consumers will use conversational systems at scale and increasingly expect personalized explanation. Finance-specific experiments also show the central risk: conversational trust can make people more likely to accept bad advice.

The 4ormlQ opportunity is therefore not to become an automated financial adviser

It is to become a private decision-readiness layer around major financial choices: a place where a person can explore without performing, understand the trade-offs, test what happens now and later, identify what they do not know, and decide what context moves into the professional conversation.

The consumer benefit is clarity and preparation. The professional benefit is better context and less reconstruction. The business benefit is a clearer record of what happened and why. Those benefits must be validated together, and none should come at the expense of autonomy, privacy or professional accountability.

The industry built systems around the transaction. 4ormlQ can build intelligence around the person before the transaction.

4orml research thesis

The next step is not a bigger claim. It is a controlled mortgage-discovery pilot with pre-registered measures for comprehension, context, professional time, privacy, pressure and overreliance.

Method, disclosure and evidence grading.

Method

This paper is a focused synthesis of published research and public deployment evidence completed in October 2026. It combines peer-reviewed studies in human-computer interaction, psychology, behavioural finance and digital health with Canadian consumer research, public-sector guidance, company operating metrics and selected customer case studies. It is not a systematic review of every paper in these fields, legal advice, a representative 4orm consumer study or evidence that the proposed 4ormIQ architecture has already produced the outcomes discussed.

EXHIBIT 30 · HOW TO WEIGH THE EVIDENCE

EVIDENCE LEVEL	USE IN THIS PAPER	TYPICAL LIMITATION
Peer-reviewed experiments and meta-analyses	Mechanisms, causal effects and pooled findings	May use hypothetical tasks or populations unlike Canadian financial consumers.
Randomized controlled trials	Comparative engagement or outcome effects	Sector transfer may be limited; product-specific.
Large observational deployments	Real-world workflow and access signals	Confounding and supplier involvement can limit causal interpretation.
Government and regulator research	Canadian consumer context, policy and design principles	Surveys report attitudes or self-reported behaviour within their scope.
Company operating metrics	Scale and adoption	Selected, self-reported and usually not independently audited.
Customer and supplier case studies	Implementation examples and hypotheses	Selection bias and attribution risk.
Testimonials	Human language and experience themes	Not representative and unsuitable for prevalence claims.
4orm analysis	Product hypotheses, architecture and research agenda	Requires validation in real use.

Source: 4orm evidence grading applied throughout this paper.

Disclosure

4orm is developing a commercial product in this area and has an interest in the conclusions. None of the organizations cited has endorsed this paper or 4orm. Proposed 4ormIQ capabilities are forward-looking product concepts, not outcomes from a live deployment. Supplier-reported metrics are identified as such. Where studies are funded by or authored by employees of the supplier being evaluated, that relationship is noted when material.

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